LANDFORMS BY RUNNING WATER / INDIA DRAINAGE AND INTERLINKING

BASIN ANATOMY → RIVER-WORK ENGINE → LONG-PROFILE CONTROLS → MEANDER-FLOODPLAIN LOOP → DRAINAGE-PATTERN MATRIX → DRAINAGE-EVOLUTION FIREWALL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◇ SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

00

STAGE 00

BASIN ANATOMY

BASIN ANATOMY · DRAINAGE DIVIDE · RUNOFF ENTERS TRIBUTARIES · TRUNK STREAM · BASIN RULE · UPSTREAM CHANGE TRAVELS DOWNSTREAM · MUST REMEMBER · FLUVIAL SYSTEMS CONNECT RAINFALL-RUNOFF, DRAINAGE-NETWORK ORGANISATION, EROSION, TRANSPORT, DEPOSITION

CORE CLAIM

- RIDGELINE → watershed / drainage divide
- SLOPES → runoff enters tributaries

MECHANISM

- TRIBUTARIES → trunk stream → outlet
- BASIN RULE → upstream change travels downstream

CONSEQUENCE / LIMIT

- MUST REMEMBER: Fluvial systems connect rainfall-runoff, drainage-network organisation, erosion, transport, deposition, graded-profile adjustment and floodplain evolution to
- Himalayan antecedent rivers, Peninsular drainage and basin-scale interlink proposals.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Himalayan antecedent rivers, Peninsular drainage and basin-scale interlink proposals.

01

STAGE 01

RIVER-WORK ENGINE

RIVER-WORK ENGINE · ENERGY + DISCHARGE · EROSION CAPACITY · LOAD SIZE + VOLUME · TRANSPORT DEMAND · COMPETENCE FALLS · DEPOSITION BEGINS · BED FORM ALTERS VELOCITY AND LOAD

ENERGY + DISCHARGE → erosion capacity → LOAD SIZE + VOLUME → transport demand → COMPETENCE FALLS → deposition begins → FEEDBACK → bed form alters velocity and load

CORE CLAIM

- ENERGY + DISCHARGE → erosion capacity
- LOAD SIZE + VOLUME → transport demand

MECHANISM

- COMPETENCE FALLS → deposition begins
- FEEDBACK → bed form alters velocity and load

CONSEQUENCE / LIMIT

- ENERGY + DISCHARGE → erosion capacity
- LOAD SIZE + VOLUME → transport demand

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FEEDBACK → bed form alters velocity and load.

02

STAGE 02

LONG-PROFILE CONTROLS

LONG-PROFILE CONTROLS · BASE LEVEL · POTENTIAL-ENERGY CHANGE · ROCK + DISCHARGE · LOCAL INCISION · ABRASION, TRANSPORT AND STORAGE · GRADED CURVE · MOVING DYNAMIC BALANCE

UPLIFT / BASE LEVEL → potential-energy change → ROCK + DISCHARGE → local incision → SEDIMENT → abrasion, transport and storage → GRADED CURVE → moving dynamic balance

CORE CLAIM

- UPLIFT / BASE LEVEL → potential-energy change
- ROCK + DISCHARGE → local incision

MECHANISM

- SEDIMENT → abrasion, transport and storage
- GRADED CURVE → moving dynamic balance

CONSEQUENCE / LIMIT

- UPLIFT / BASE LEVEL → potential-energy change
- ROCK + DISCHARGE → local incision

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SEDIMENT → abrasion, transport and storage; GRADED CURVE → moving dynamic balance.

03

STAGE 03

MEANDER-FLOODPLAIN LOOP

MEANDER-FLOODPLAIN LOOP · OUTER BANK · FASTER FLOW · INNER BANK · SLOWER FLOW · POINT BAR · NECK CUTOFF · OX-BOW LAKE

OUTER BANK → faster flow → erosion → INNER BANK → slower flow → point bar → NECK CUTOFF → ox-bow lake → OVERBANK SILT → floodplain + natural levee

CORE CLAIM

- OUTER BANK → faster flow → erosion
- INNER BANK → slower flow → point bar

MECHANISM

- NECK CUTOFF → ox-bow lake
- OVERBANK SILT → floodplain + natural levee

CONSEQUENCE / LIMIT

- OUTER BANK → faster flow → erosion
- INNER BANK → slower flow → point bar

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INNER BANK → slower flow → point bar.

04

STAGE 04

DRAINAGE-PATTERN MATRIX

DRAINAGE-PATTERN MATRIX · UNIFORM RESISTANCE · ALTERNATING RESISTANT BANDS · JOINTS AND FAULTS · OUTWARD HIGH · INWARD BASIN

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
DENDRITIC → uniform resistance	RECTANGULAR → joints and faults	DENDRITIC → uniform resistance
TRELLIS → alternating resistant bands	RADIAL / CENTRIPETAL → outward high / inward basin	TRELLIS → alternating resistant bands

CORE CLAIM

- DENDRITIC → uniform resistance
- TRELLIS → alternating resistant bands

MECHANISM

- RECTANGULAR → joints and faults
- RADIAL / CENTRIPETAL → outward high / inward basin

CONSEQUENCE / LIMIT

- DENDRITIC → uniform resistance
- TRELLIS → alternating resistant bands

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RADIAL / CENTRIPETAL → outward high / inward basin.

05

STAGE 05

DRAINAGE-EVOLUTION FIREWALL

DRAINAGE-EVOLUTION FIREWALL · RIVER OLDER THAN UPLIFT · COVER PATTERN INHERITED BELOW · HEADWARD EROSION DIVERTS NEIGHBOUR · FIELD CUES · ELBOW-WIND GAP

ANTECEDENT → river older than uplift → SUPERIMPOSED → cover pattern inherited below → CAPTURE → headward erosion diverts neighbour → FIELD CUES → gorge / discordance / elbow-wind gap

CORE CLAIM

- ANTECEDENT → river older than uplift
- SUPERIMPOSED → cover pattern inherited below

MECHANISM

- CAPTURE → headward erosion diverts neighbour
- FIELD CUES → gorge / discordance / elbow-wind gap

CONSEQUENCE / LIMIT

- ANTECEDENT → river older than uplift
- SUPERIMPOSED → cover pattern inherited below

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FIELD CUES → gorge / discordance / elbow-wind gap.

06

STAGE 06

INDIA RIVER-SYSTEM CONTRAST

INDIA RIVER-SYSTEM CONTRAST · HIGH RELIEF, MIXED PERENNIAL INPUTS · OLD SHIELD, MONSOON-DOMINANT REGIME · LONG RIVERS AND MAJOR DELTAS · NARMADA-TAPI TROUGHS + SHORT GHATS STREAMS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
HIMALAYAN → high relief, mixed perennial inputs	EAST → long rivers and major deltas	HIMALAYAN → high relief, mixed perennial inputs
PENINSULAR → old shield, monsoon-dominant regime	WEST → Narmada-Tapi troughs + short Ghats streams	PENINSULAR → old shield, monsoon-dominant regime

CORE CLAIM

- HIMALAYAN → high relief, mixed perennial inputs
- PENINSULAR → old shield, monsoon-dominant regime

MECHANISM

- EAST → long rivers and major deltas
- WEST → Narmada-Tapi troughs + short Ghats streams

CONSEQUENCE / LIMIT

- HIMALAYAN → high relief, mixed perennial inputs
- PENINSULAR → old shield, monsoon-dominant regime

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WEST → Narmada-Tapi troughs + short Ghats streams.

07

STAGE 07

SOURCE-COURSE-MOUTH RAIL

SOURCE-COURSE-MOUTH RAIL · TIBETAN HEADWATERS · ARABIAN SEA · DEVPRAYAG NAME · BENGAL DELTA · JOINED DELTA

INDUS → Tibetan headwaters → Arabian Sea

GANGA → Devprayag name → Bengal delta → BRAHMAPUTRA → Tsangpo-Siang → joined delta → PENINSULA → Godavari-Krishna-Kaveri / Narmada-Tapi

CORE CLAIM

- INDUS → Tibetan headwaters → Arabian Sea
- GANGA → Devprayag name → Bengal delta

MECHANISM

- BRAHMAPUTRA → Tsangpo-Siang → joined delta
- PENINSULA → Godavari-Krishna-Kaveri / Narmada-Tapi

CONSEQUENCE / LIMIT

- INDUS → Tibetan headwaters → Arabian Sea
- GANGA → Devprayag name → Bengal delta

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BRAHMAPUTRA → Tsangpo-Siang → joined delta; PENINSULA → Godavari-Krishna-Kaveri / Narmada-Tapi.

08

STAGE 08

DELTA BUDGET

DELTA BUDGET · BASIN SEDIMENT DELIVERY · WAVES, TIDES AND CYCLONE SCOUR · COMPACTION + RELATIVE SEA LEVEL · DISTRIBUTARY MOBILITY OR CONFINEMENT

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
INPUT → basin sediment delivery	VERTICAL → compaction + relative sea level	INPUT → basin sediment delivery
REMOVAL → waves, tides and cyclone scour	CHANNELS → distributary mobility or confinement	REMOVAL → waves, tides and cyclone scour

CORE CLAIM

- INPUT → basin sediment delivery
- REMOVAL → waves, tides and cyclone scour

MECHANISM

- VERTICAL → compaction + relative sea level
- CHANNELS → distributary mobility or confinement

CONSEQUENCE / LIMIT

- INPUT → basin sediment delivery
- REMOVAL → waves, tides and cyclone scour

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERTICAL → compaction + relative sea level; CHANNELS → distributary mobility or confinement.

09

STAGE 09

TRANSFER VIABILITY TEST

TRANSFER VIABILITY TEST · SEASONAL WATER AFTER EXISTING CLAIMS · DEMAND, SOIL, AQUIFER AND DELIVERY FIT · LOSS, ENERGY, ECOLOGY AND PEOPLE · RECHARGE, EFFICIENCY, REUSE, OPERATIONS · CLOSE DISTINCTION · V-SHAPED VALLEY, GORGE, WATERFALL, MEANDER, OXBOW, LEVEE, DELTA AND

DONOR → seasonal water after existing claims

RECIPIENT → demand, soil, aquifer and delivery fit → CORRIDOR → loss, energy, ecology and people → ALTERNATIVES → recharge, efficiency, reuse, operations

CORE CLAIM

- DONOR → seasonal water after existing claims
- RECIPIENT → demand, soil, aquifer and delivery fit

MECHANISM

- CORRIDOR → loss, energy, ecology and people
- ALTERNATIVES → recharge, efficiency, reuse, operations

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: V-shaped valley, gorge, waterfall, meander, oxbow, levee, delta and alluvial fan occupy different energy and valley settings
- drainage pattern differs from drainage basin, and river capture differs from planned diversion.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: V-shaped valley, gorge, waterfall, meander, oxbow, levee, delta and alluvial fan occupy different energy and valley settings; drainage pattern differs from drainage basin, and river capture differs from planned diversion.

10

STAGE 10

KEN-BETWA BOUNDARY

KEN-BETWA BOUNDARY · KEN BASIN TO BETWA BASIN · OFFICIAL MATERIAL SAYS UNDER IMPLEMENTATION · PANNA CONNECTIVITY + DOWNSTREAM FLOW · STATUS IS NOT MEASURED OUTCOME · EVIDENCE LIMIT · INTERLINKING EFFECTS ARE BASIN- AND DESIGN-SPECIFIC · TRANSFER VOLUME, SEASONALITY, ENVIRONMENTAL FLOWS, SEDIMENT, DISPLACEMENT, FEDERAL CONSENT

DIRECTION → Ken basin to Betwa basin

STATUS → official material says under implementation → LANDSCAPE → Panna connectivity + downstream flow → RULE → status is not measured outcome

CORE CLAIM

- DIRECTION → Ken basin to Betwa basin
- STATUS → official material says under implementation

MECHANISM

- LANDSCAPE → Panna connectivity + downstream flow
- RULE → status is not measured outcome

CONSEQUENCE / LIMIT

- EVIDENCE LIMIT: Interlinking effects are basin- and design-specific: transfer volume, seasonality, environmental flows, sediment,
- displacement, federal consent and climate uncertainty must qualify both drought-proofing and ecological-damage claims.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Interlinking effects are basin- and design-specific: transfer volume, seasonality, environmental flows, sediment, displacement, federal consent and climate uncertainty must qualify both drought-proofing and ecological-damage claims.

11

STAGE 11

BASIN ANSWER SPINE

BASIN ANSWER SPINE · DEFINE + MAP · BASIN AND DIRECTION · PROCESS, TIMING AND SEDIMENT · BENEFIT MECHANISM AGAINST LIMITS · SELECTIVE, ADAPTIVE AND BASIN-WIDE

DEFINE + MAP → basin and direction → EXPLAIN → process, timing and sediment → ASSESS → benefit mechanism against limits → VERDICT → selective, adaptive and basin-wide

CORE CLAIM

- DEFINE + MAP → basin and direction
- EXPLAIN → process, timing and sediment

MECHANISM

- ASSESS → benefit mechanism against limits
- VERDICT → selective, adaptive and basin-wide

CONSEQUENCE / LIMIT

- DEFINE + MAP → basin and direction
- EXPLAIN → process, timing and sediment

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ASSESS → benefit mechanism against limits; VERDICT → selective, adaptive and basin-wide.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · THE INDUS, GANGA AND BRAHMAPUTRA SYSTEMS CROSS INTERNATIONAL BOUNDARIES · ANTECEDENT DRAINAGE · RIVERS PREDATING THE HIMALAYA THAT MAINTAINED THEIR COURSE BY · RIVER CAPTURE (PIRACY) · A MORE VIGOROUS RIVER, THROUGH HEADWARD EROSION, BEHEADS A · INDO-BRAHM HYPOTHESIS (KHULLAR) · A MIGHTY FORMER RIVER ALONG THE HIMALAYAN FOREDEEP WAS

OPTIONAL SOURCE DEPTH

- The Indus, Ganga and Brahmaputra systems cross international boundaries and are mostly perennial.
- Antecedent drainage: rivers predating the Himalaya that maintained their course by down-cutting as the range uplifted (deep transverse gorges).
- River capture (piracy): a more vigorous river, through headward erosion, beheads a neighbouring river — a key process in Himalayan drainage evolution.
- Indo- Brahm hypothesis (Khullar): a mighty former river along the Himalayan foredeep was

ADVANCED DEBATE

- dismembered by successive captures into today's Indus, Ganga and Brahmaputra systems.
- Most large Peninsular basins drain east; Narmada and Tapi flow west through structurally controlled
- Objective: transfer water within Bundelkhand for irrigation, drinking water and power; quote
- Concerns: submergence and fragmentation around Panna Tiger Reserve, downstream-flow effects,

LEGACY / NUANCE

- Institutions: Central Water Commission, Ken-Betwa Link Project Authority; other links for comparison — Par-Tapi-Narmada, Damanganga-Pinjal.
- Himalayan systems are mostly perennial; Peninsular regimes are more monsoon-seasonal and occupy an older landscape.
- Antecedent river = older than the mountain it cuts across (deep gorge).
- River capture driven by headward erosion (Khullar: Indo- Brahm captures).

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels